Stroke Prediction

Project report: patient data, four classifiers, and why accuracy can mislead

Notebook: Heart-stroke-detection-final.ipynb | Author: Yash Jain

1. Introduction

Stroke is one of the leading causes of death worldwide. If basic health information could flag the patients at higher risk, it would help doctors decide who needs closer attention. This project asks how well common health details (age, hypertension, heart disease, glucose level, BMI, smoking status and a few lifestyle fields) predict whether a patient has had a stroke.

I used a public Kaggle dataset of 5,110 patients, explored it, and trained four standard classifiers: logistic regression, a support vector machine, a decision tree and k-nearest neighbours. The headline result looked excellent, with 93.9% accuracy for three of the models. The more important finding is that those three models never predicted a stroke, and most of this report is about why that happened and what it means.

This is a learning project. It is not a medical tool, and nothing in it should be used to make health decisions.

2. Data

The dataset is the Stroke Prediction Dataset on Kaggle (by fedesoriano). It has 5,110 rows and 12 columns. The target is *stroke*, which is 1 if the patient had a stroke. The other columns are:

Column	Description
id	Identifier (dropped)
gender	Female 2,994, Male 2,115, Other 1
age	0.08 to 82 years (mean 43.2)
hypertension, heart_disease	0 or 1 (9.7% and 5.4% of patients)
ever_married	Yes 3,353, No 1,757
work_type	Private 2,925, Self-employed 819, children 687, Govt_job 657, Never_worked 22
Residence_type	Urban 2,596, Rural 2,514
avg_glucose_level	55 to 272 (mean 106.1)
bmi	10.3 to 97.6 (mean 28.9), 201 missing
smoking_status	never smoked 1,892, Unknown 1,544, formerly smoked 885, smokes 789
stroke	1 for 249 patients (4.9%), 0 for 4,861

The data is heavily imbalanced: only 1 patient in 20 had a stroke. The BMI column is the only one with missing values, and it has some very high values (the maximum is 97.6) that look like errors or extreme cases. "Unknown" in smoking status means the information was not available, and it is a

large group.

3. Preprocessing

Missing values. The 201 missing BMI values were filled with the most common value in the column (28.7).

Encoding. I converted the text columns to numbers by hand: ever_married (Yes 1, No 0), gender (Female 0, Male 1, Other 2), Residence_type (Rural 0, Urban 1), smoking_status (formerly smoked 0, never smoked 1, smokes 2, Unknown 3) and work_type (Private 0, Self-employed 1, children 2, Govt_job 3, Never_worked 4). These numbers have no real order, but the models treat them as if they do.

Age groups. After looking at the correlations, I replaced age with five bands: 0 to 12, 13 to 19, 20 to 30, 31 to 60 and 61 to 100.

Split. 80% training and 20% test with random_state 42 and no stratification. The test set has 1,022 patients, of whom 62 had a stroke (the training set has 187). No scaling was applied.

4. Exploratory analysis

Correlations. Before binning age, the correlations with stroke were: age 0.245, heart disease 0.135, average glucose level 0.132, hypertension 0.128, ever married 0.108, BMI 0.038 and a few small negative ones for smoking status (-0.066) and work type (-0.032). The signs for smoking status and work type do not mean much because their codes are arbitrary. Age is the strongest by a good margin, and nothing is above 0.25, which tells us this will be a hard prediction problem.

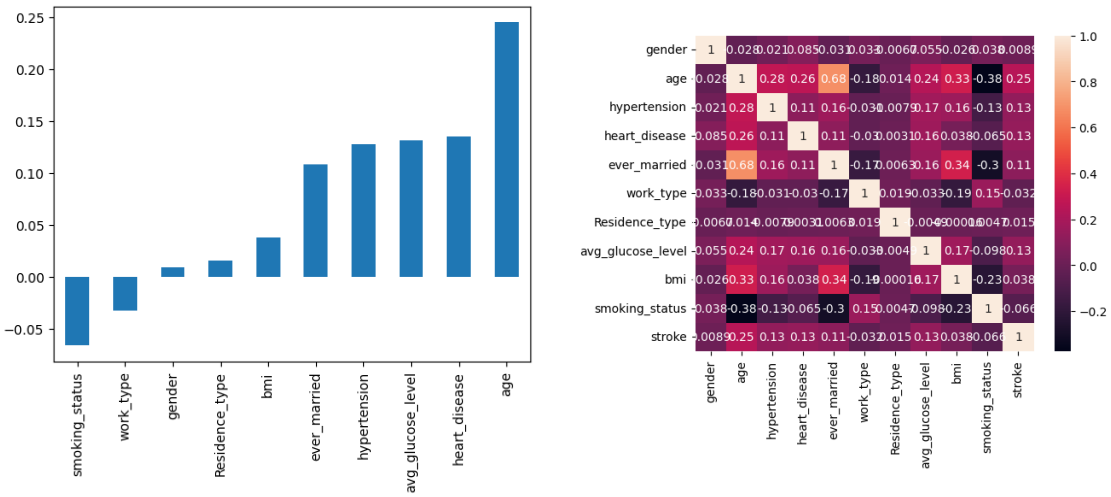


Figure 1. Correlation of each feature with stroke (left) and the full correlation matrix (right).

Count plots and why they mislead. My notebook drew a grid of count plots comparing each feature with stroke. A count plot shows how many patients are in each group, not how likely each group is to have a stroke. Because most patients do not have hypertension or heart disease, more of the strokes in absolute terms are among patients without those conditions. I read this in the notebook as "people with a lower chance of hypertension and heart disease have an increased chance of stroke", which is wrong. The same problem applies to my comment that non-smokers are more likely to have a stroke than smokers (there are simply more non-smokers), and to the comment that people with a BMI between 20 and 50 are at higher risk (that is where most patients are).

What the rates say. To correct this, I recomputed the stroke rate inside each group from the CSV (Figure 2). Age dominates: 13.6% of patients aged 61 to 100 had a stroke, against 3.1% for ages 31 to 60 and under 0.5% for everyone younger. Hypertension raises the rate from 4.0% to 13.3%, and heart disease from 4.2% to 17.0%. Smoking is less clear: former smokers have the highest rate (7.9%), current smokers 5.3% and never-smokers 4.8%. The Unknown group has only 3.0%, but 35% of it are children aged 12 or under, compared with 11.5% of all patients, and Unknown adults have a rate of 5.2%.

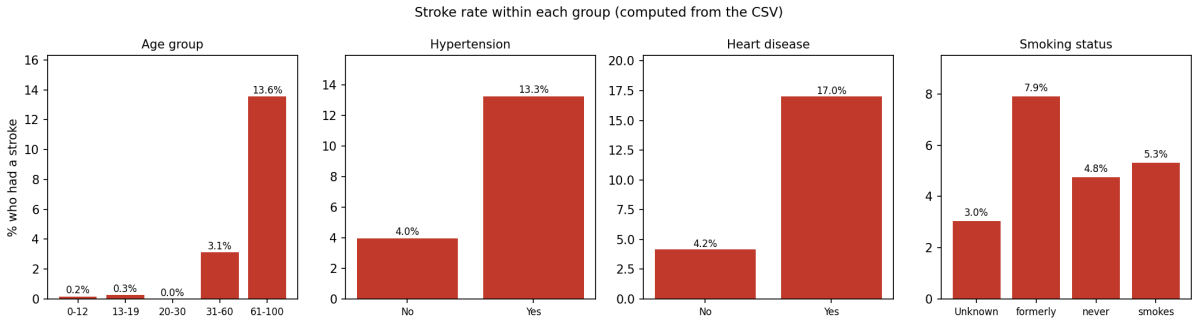


Figure 2. Stroke rate within each group, computed from the CSV (not part of the original notebook).

Married patients (6.6% against 1.7%) and self-employed patients (7.9%, against about 5% for private and government jobs) also have higher rates, but both are tangled up with age. Patients with an average glucose level above 125 have a stroke rate of about 10%, against under 4% for lower levels. Gender and residence type make almost no difference (4.7% against 5.1% for gender, and 4.5% against 5.2% for rural against urban).

Missing BMI. The 201 patients with a missing BMI had a stroke rate of about 20%, four times the average. Filling all of them with one value hides this, and it may be informative in itself.

5. Modelling

Four classifiers from scikit-learn were trained on the training set with their default settings: LogisticRegression, SVC, DecisionTreeClassifier and KNeighborsClassifier (5 neighbours). Nothing was tuned, scaled or class-weighted. The decision tree was created without a random_state, so its results can change between runs.

6. Results

	Logistic regression	SVM	Decision tree	KNN
Accuracy	0.9393	0.9393	0.9110	0.9374
F1 (stroke class)	0.000	0.000	0.195	0.000
Strokes found (of 62)	0	0	11	0
False alarms	0	0	40	2
Strokes missed	62	62	51	62
MAE / MSE	0.0607	0.0607	0.0890	0.0626

Logistic regression and the SVM have identical results: both predicted "no stroke" for every one of the 1,022 test patients. KNN made two "stroke" predictions, both wrong. The decision tree is the only

model that found any strokes (11 of 62), at the cost of 40 false alarms, and it has the lowest accuracy of the four.

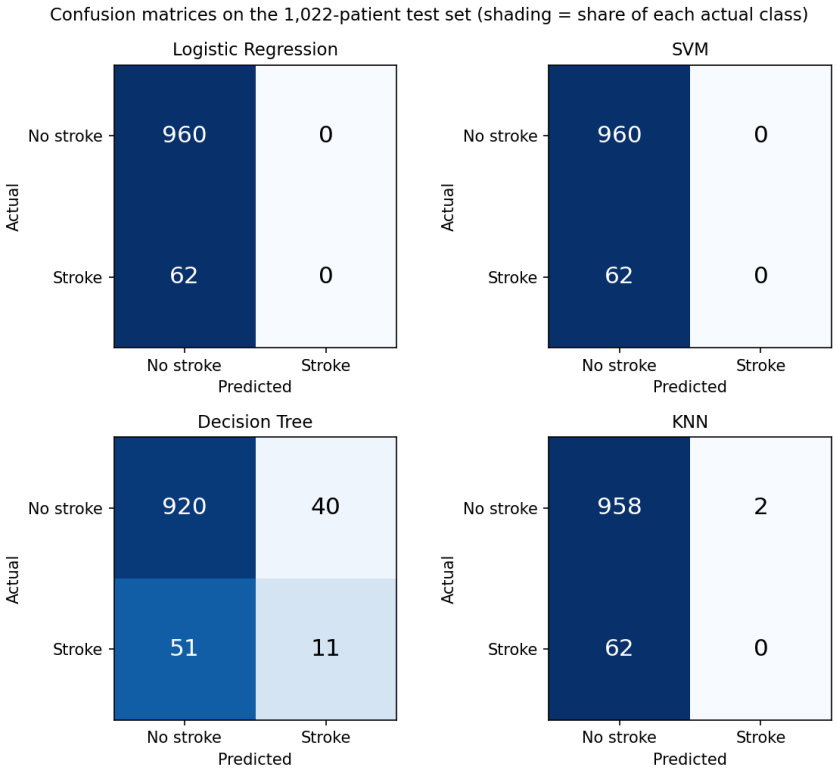


Figure 3. Confusion matrices for the four models on the test set (rows are actual, columns are predicted).

Why the accuracy is misleading. In the test set 960 of the 1,022 patients (93.9%) did not have a stroke. A "model" that always answers "no stroke" gets exactly that accuracy, which is what logistic regression and the SVM did. So the 93.9% says nothing about whether the model has learned anything. What matters here is the stroke class, where the three models have an F1 score of 0 and the tree a score of 0.19 (precision 0.22, recall 0.18).

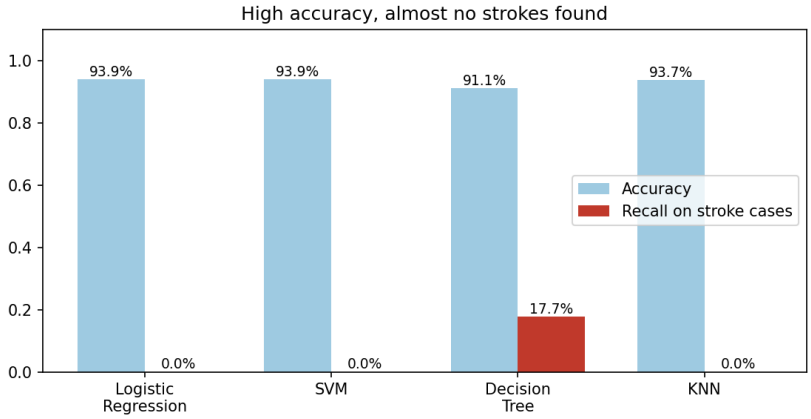


Figure 4. Accuracy against recall on stroke cases. The notebook's own bar chart showed only the accuracy.

The decision tree results change between runs. When I re-ran the same preprocessing, I got 90.6% accuracy and 14 strokes found with 48 false alarms, not 91.1% and 11 with 40. The notebook output is what the table above reports.

7. Limitations

- **The wrong metric.** The notebook compares models by accuracy. With 94% of the test set being the same class, accuracy cannot separate a useful model from one that never predicts a stroke. The F1 score of 0.0 was in my output, and I did not follow it up.
- **The conclusion is not supported.** It says all four models are about equally good, "so, we can use any of these models to predict the heart stroke". Three of them never predict a stroke. It also quotes 93.8% for three models and 91.8% for the tree, while the outputs show 93.9%, 93.9%, 93.7% and 91.1%.
- **Misread charts.** The statements about hypertension, heart disease, smoking and BMI came from count plots (section 4). The rates show the opposite for hypertension and heart disease.
- **Class imbalance was ignored.** There was no stratified split, no class weights, no oversampling and no change to the decision threshold.
- **Weak preprocessing.** The BMI fill with a constant hides a group with a high stroke rate. Smoking and work type were label-encoded as if they were ordered. No scaling was applied for logistic regression, SVM and KNN, which depend on it (logistic regression also gives a convergence warning when I run it). Age was turned into five bands, which throws away information from the strongest feature.
- **Metrics that do not fit.** For 0/1 predictions the MAE and MSE are the same number (the error rate). The log loss was computed on hard predictions, not probabilities, so the values around 2.2 to 3.2 are not meaningful.
- **Unstable decision tree.** No random_state, so results vary from run to run.
- **A single split and no tuning.** The test set has only 62 stroke cases, so one patient moves the recall by about 1.6 percentage points.
- **Not a clinical dataset.** I do not know how the data was collected, what "stroke" covers or how patients were selected, so the results should not be generalised.

8. Conclusion

Age is by far the strongest factor in this data, followed by heart disease, hypertension and a high glucose level. Smoking and the lifestyle fields matter less once age is taken into account. Four default classifiers all end up with accuracy between 91% and 94%, but this is almost entirely due to the 95% of patients who did not have a stroke. Logistic regression, SVM and KNN effectively never predict a stroke, and the decision tree finds about one in five at the cost of many false alarms.

So the honest result of this project is that these models, in this setup, cannot predict stroke. The useful part is the analysis of what went wrong, and a clear list of things to change.

9. Future work

- Evaluate with recall, precision, F1, ROC AUC and precision-recall curves, and report predicted probabilities, not just labels.

- Use a stratified split and cross-validation, and handle the imbalance with class weights, SMOTE or threshold tuning.
- Keep age as a number, one-hot encode the categorical columns, scale the inputs and add a flag for missing BMI (using the median to fill it).
- Try a random forest and gradient boosting, and tune the main hyperparameters.
- Redo the exploratory analysis with rates rather than counts.
- Set random_state on every model so the results can be reproduced.
- Fix the conclusion and the chart comments in the notebook so they match the results.

10. Tools used

Python, pandas, NumPy, Matplotlib, Seaborn, scikit-learn, Google Colab.